MIL-C-43609B
13 June 1977
SUPERSEDING
MIL-C-43609A
29 March 1974

MILITARY SPECIFICATION

CABINETS, FOOD WARMING, ELECTRICALLY HEATED, DRAWER TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one size of countertop food warming cabinet and one size mobile food warming cabinet.

1.2 Classification. Food warming cabinets shall be of the following types and styles as specified (see 6.2):

Type I - For countertop use (2 drawers)

Style L - Leg mounted

Type II - Floor model with (3 drawers)

Style L - Leg mounted
Style C - Caster mounted

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U. S. Army Natick Research and Development Command, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 7310

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SPECIFICATION

FEDERAL

FF-C-77 - Casters, Rigid and Swivel (Institutional Duty)

MILITARY

MIL-B-2316 - Bakery Equipment, Including Units, Assemblies, Repair Parts and Tools, Preparation for Delivery of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

UNDERWRITERS LABORATORIES INC. (UL)

UL 197 - Commercial Electric Cooking Appliances

(Application for copies should be addressed to the Underwriters Laboratories Inc., 207 East Ohio Street, Chicago, IL 60611; 333 Pfingsten Road, Northbrook, IL 60062; 1285 Walt Whitman Road, Melville, L.I. NY 11746; 1655 Scott Boulevard, Santa Clara, CA 95050, or 2602 Tampa East Boulevard, Tampa, FL 33619.)

NATIONAL SANITATION FOUNDATION (NSF)

Standard No. 4 - Commercial Cooking and Hot Food Storage Equipment

Listing of Food Service Equipment

(Copies may be obtained from the National Sanitation Foundation; NSF Building, Ann Arbor, MI 48105.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 167 - Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip
A 176 - Stainless and Heat-Resisting Chromium Steel Plate, Sheet, and Strip

(Copies may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

C 73 - Dimensions of Attachment Plugs and Receptacles

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10019.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.)

3. REQUIREMENTS

* 3.1 Standard product. The cabinet shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product with any added features needed to comply with the requirements of this specification. Modifications to add features shall not incorporate different parts unless such parts are used on other current commercial models. Standard or modified commercial products furnished in accordance with this specification shall be identifiable by all regular manufacturer's or commercial service organizations servicing the brand involved. Service organizations shall be capable of providing complete parts and repair services on models furnished to the Government consistent with their normal commercial practices.

3.2 Codes and standards. The cabinets shall comply with the applicable requirements of UL 197 and NSF Standard No. 4.

3.2.1 Compliance. Prior to commencing production, the contractor shall submit to the contracting officer or his authorized representative, satisfactory evidence that the cabinets he proposes to furnish under this specification meet the applicable requirements of UL and NSF as follows:

3.2.1.1 UL. Acceptable evidence of meeting the requirements of UL shall be the UL label, a UL listing mark, or a certified test report from a recognized independent testing laboratory acceptable to the Government, indicating that the cabinets have been tested and conform to UL 197.

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3.2.1.2 NSF. Acceptable evidence of meeting the requirements of NSF shall be:

(1) A listing in the current edition of the NSF "Listing of Food Service Equipment" and display of the NSF seal on the finished cabinets, or

(2) A certification for the cabinets issued by NSF under their special one-time contract evaluation/certification service, or

(3) A certified test report from a recognized independent testing laboratory acceptable to the medical department of the service for which the cabinets are being procured, indicating that the cabinets have been tested and conform to NSF Standard No. 4.

3.3 Materials. Materials not definitely specified shall be of the type and quality normally furnished on the manufacturer's standard food warming cabinets.

3.3.1 Stainless steel. Stainless steel shall conform to the chemical composition of types 302 or 304 as specified in ASTM A 167 or type 430 as specified in ASTM A 176.

3.4 Design and construction. Food warming cabinets shall be designed to warm rolls and other foods and maintain them at serving temperatures. Type I cabinets shall have two heated drawers and be for countertop use. Type II cabinets shall have three heated drawers and be mounted on four casters for style C or four legs for style L, as applicable. Cabinet design shall provide for circulation of thermostatically controlled heat around each drawer and shall have a separate, manual, adjustable moisture control for each drawer. The moisture control shall operate a moisture retaining cover which seals the drawer for humid storage when closed and for dry storage allows moisture to escape when retaining cover is fully open. Each drawer shall be capable of holding a full size 4-1/2 inch deep by 12-inch by 20-inch (114 by 305 by 508 mm) cafeteria pan and shall have a minimum capacity of 7 dozen 6-inch (152 mm) long hot dog rolls when tested as specified in 4.4.1. Heating elements shall be capable of heating drawers from room temperature to 150°F (66°C) at a minimum average rate of 3°F (1.6°C) temperature rise per minute (see 4.4.2), and shall maintain drawer temperatures of 150°F and 200°F (66°C and 93°C) within plus or minus 10°F (55°C) with thermostat set at 150°F and 200°F (66° and 93°C), respectively (see 4.4.3). Type I cabinets shall be not greater than 29-1/2 inches (749 mm) wide and 22 inches (559 mm) deep by 23 inches (584 mm) high. Type II cabinets shall be not greater than 35 inches wide by 25 inches (889 by 635 mm) deep, excluding bumpers, by 37 inches (940 mm) high.

3.4.1 Cabinet body. Outer top, front, back, and sides shall be stainless steel specified in 3.3.1. Outer bottom shall be stainless steel specified in 3.3.1, aluminized steel, coated carbon steel to resist corrosion or galvanized steel. Inner top, back, and sides shall be stainless steel specified in 3.3.1, galvanized steel, or aluminized steel. Minimum metal thickness shall be as shown in table I. The space between the inner and outer walls shall be in-

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fiber, installed to prevent settling (see 4.2.2.1). Type II cabinets shall have a replaceable rubber or urethane bumper on all four sides of the base and a rubber-bumpered handle shall be welded or bolted to the outer right side of the body.

TABLE I. Metal thickness (minimum)

Component	Gauge	Thickness inches	(mm)
Outer and inner top, back, sides, front and inner bottom	24	0.024	(0.61)
Outer bottom	20	0.034	(0.86)
Drawer front	24	0.024	(0.61)
Drawer pan	24	0.024	(0.61)
Drawer runners	16	0.060	(1.5)

3.4.2 Drawers. Drawer front and drawer pan shall be stainless steel specified in 3.3.1 except drawer pan material shall be limited to types 302 or 304. Minimum thickness of drawer components fabricated of stainless steel shall be as shown in table I. Other drawer parts which do not come in contact with food shall be stainless steel or nickel-plated or chrome-plated steel except the handle may be a replaceable heat-resistant rubber or plastic handle. Drawers shall slide on runners provided with nylon wheels, self-lubricating sintered bearings, or ball or needle bearings. Stops shall be provided to prevent accidentally pulling drawers all the way out. Drawers shall fit flush and tight with cabinet exterior. The drawer pan shall be formed of one piece of material.

* 3.4.3 Casters. Type II, style C cabinets shall have four casters conforming to class B, wheel diameter 5 inch (127 mm), style b, mounting A, Nn P, or T of FF-C-77. The cabinet four corners shall have provisions for casters to be bolted to the cabinet base. Two type I (rigid) casters shall be located at the left end (when viewing cabinet front) and two type II (swivel) casters with locking features shall be located at the right end.

3.4.4 Electrical requirements. Unless otherwise specified (see 6.2), cabinets shall be designed for operation on a nominal 120-volt, 60-hertz, 1-phase, system.

3.4.5 Heating elements. Cabinets shall have heating elements with a minimum total wattage of 800 for type I and 1000 for type II, except that when individual elements are provided for each drawer, the minimum wattage shall be 400 per element.

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* 3.4.6 Thermostat. There shall be one thermostat for regulating the temperature throughout the cabinet or there shall be individual thermostats for each drawer. Thermostats shall be of the snap action type. Thermostats shall have an integral "ON-OFF" switch and shall have a low setting of not less than 145°F (63°C), and an adjustable range up to at least 200°F (93°C). The "OFF" position and temperature dial settings shall be clearly marked on the thermostat knob. Thermostats shall have a signal light to indicate when the elements are energized. Thermostat bulb shall be positioned so it is protected from possible damage during operation.

* 3.4.7 Electric cord. An electric UL approved, oil-resistant-type service cord (types SO, STO, SJO, SJTO, HSO or HSJO) shall be permanently connected at the cabinet. Cord shall have a grounding conductor and shall have a three-prong attachment plug cap in accordance with ANSI C 73 for connection to the supply service. Cord length shall be not less than 3 feet (91 cm) nor more than 6-feet (183 cm).

* 3.5 Marking. Identification shall be permanently marked directly on the item or on an aluminum, brass, or stainless steel plate permanently affixed to the item. Marking shall be stamped, embossed, engraved or applied by photosensitive means. The plate thickness shall be not less than 0.012 inch (0.3 mm).

* 3.5.1 Marking for identification. Each cabinet shall be permanently and legibly marked with the manufacturer's model and name, trade name or trademark easily identified with said manufacturer and National stock number, contract number, serial number of unit and electrical rating. In addition, such information as required by UL 197 for the cabinet shall be included on the plate.

* 3.6 Workmanship. All components and assemblies of the cabinet shall be free from dirt and other extraneous material, burrs, slivers, rough die, tool and grind marks, dents and cracks. Castings and molded parts if used shall be free of sand, fins, pits, blow holes and sprues. External surfaces shall be free from sharp edges and sharp corners.

* 3.6.1 Metal fabrication. Metal used in the fabrication on the cabinet shall be free from kinks. Forming and shearing shall not cause damage to the metal.

* 3.6.2 Welding and brazing. The surfaces of parts to be welded or brazed shall be free from rust, scale, paint, grease and other foreign matter. Welds shall be continuous, smooth and free of cracks, burn holes, undercuts, fissures or incomplete fusion. All scale shall be removed from the finish weld area.

* 3.6.3 Fastening devices. Threaded fasteners and rivet holes shall have burrs removed. Threaded fasteners shall not be broken, cracked or stripped and shall be drawn tight. Rivets shall fill the hole completely and the heads shall be in full contact with the surface of the member and concentric with the hole.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless other specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.2.2 In-process inspection.

4.2.2.1 In-process examination. Installation of insulation shall be examined for compliance with 3.4.1. Particular attention shall be given to type of insulation and method of application Whenever nonconformance is noted, correction shall be made to the process and all items processed.

4.2.3 End item inspection. The inspection lot shall consist of all food warming cabinets of one type and style offered for inspection at one time.

4.2.3.1 Visual examination. The end item shall be examined for defects in table II. The inspection level shall be II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE II. Classification of visual defects

Examine	Defect	Classification Major	Minor
Standard product	Cabinet not in accordance with manufacturers standard product except for changes necessary to comply with specification requirements.	X	

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TABLE II. Classification of visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Construction and workmanship (general)	Any component missing.	X	
	Any component fractured, or malformed.	X	
	Any wearing part not accessible for adjustment or replacement.	X	
	Any component misplaced or not in alignment.	X	
	Any burr, sliver, sharp edge, rough die, tool, gouge, trim or grind marks.	X	
	Any component fractured, buckled, bent, dented, punctured or malformed.	X	
	Functioning component that is inoperative or will not operate as intended.	X	
	Functioning component that required abnormal force to operate.	X	
	Any component not in proper alignment.	X	
	Any component loose, other than threaded fastener.		X
	Any adjustable assembly that cannot be adjusted.	X	
	Any adjustable assembly not properly adjusted to perform the intended function.		X
Construction (detailed) body	Rubber not replaceable.	X	
Drawers	Drawer slides not provided with bearings as specified.	X	
	Handle not as specified.	X	
	Moisture control not provided on each drawer.	X	
	Drawer binds or is loose when withdrawn.		X

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TABLE II. Classification of visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Drawers (cont'd)	Stop not provided to prevent accidental pulling of drawer all the way out.	X	
Casters	Not correct type.	X	
	Not located as specified.	X	
Thermostat	Not correct operating range.	X	
	Signal light not provided or does not operate as specified.	X	
	Knob binds or is loose.	X	
	Dial markings not as specified or illegible.	X	
Cord	Not provided with 3-prong grounding type plug cap.	X	
	Not type specified.	X	
Welding or brazing (as applicable)	Not free from rust, scale, paint, grease and other foreign matter.	X	
	Welds not continuous, smooth and free of cracks, burn holes, undercuts, fissures or incomplete fusion.	X	
	Finished weld area shows evidence of scale.	X	
	Flux not removed.	X	
Fastening devices			
Threaded fasteners	Missing, broken, stripped or fractured.	X	
	Loose.	X	
Rivets	Loose, not in full contact with members riveted.	X	
Identification	Missing, incomplete, or illegible.	X	

4.2.3.2 Dimensional examination. Dimensional examination shall be made of the completed food warming cabinets to determine conformance with dimensions specified. Any dimension not within the specification tolerance shall be a defect. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of defects per hundred units.

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4.2.3.3 End item testing. One item selected at random from the first produced lot shall be tested as specified in 4.4.1, 4.4.2 and 4.4.3. Failure of any test shall be cause for rejection of the first produced cabinet. Sample cabinets shall be tested as specified in 4.4.3. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units.

4.2.4 Inspection of packaging. Preservation, packaging, packing and marking shall be inspected in accordance with the quality assurance provisions in MIL B-2316.

4.3 Certificate examination. Certificates of compliance, certified test reports, approval labels or listing marks for codes and standards, as applicable, that are submitted as proof of compliance in accordance with 3.2.1.1, 3.2.1.2 and 3.4.5, shall be examined and validated (see 6.3).

4.4 Test methods.

4.4.1 Drawer capacity. The drawers shall be loaded with hot dog rolls to determine conformance with the minimum capacity specified in 3.4. The rolls shall be removed and a pan conforming to the dimensions specified in 3.4 shall be placed in each drawer to determine conformance to 3.4. Nonconformance to the requirements of 3.4 shall constitute failure of this test.

4.4.2 Heat-up time. The cabinet shall be connected to the type electrical source specified (see 3.4.4). A thermocouple shall be located in the center of each drawer and connected to an external indicating instrument, and the drawer closed. The moisture control shall be closed to retain the food moisture. The thermostat or thermostats (on stacked single drawer units) shall be set at 150°F (66°C). The average temperature rise per minute per drawer from the time of thermostat setting until 150°F (66°C) is reached shall be checked to determine conformance with the heat-up time requirements of 3.4. Nonconformance with the requirements of 3.4 shall constitute failure of this test.

4.4.3 Temperature maintainability. With each drawer thermostat set at 150°F (66°C), the cabinet shall be operated until the thermostat has cycled off three times. Each drawer temperature reading shall be taken at 10-minute intervals for a minimum of 1-hour. Each thermostat shall then be set at 200°F (93°C) and the procedure repeated. Failure to maintain the temperature in each drawer within the limits specified in 3.4 shall constitute failure of this test.

5. PACKAGING

5.1 Preservation-packaging, packing and marking. Each complete cabinet shall be preserved-packaged, packed and marked in accordance with the applicable requirements of MIL-B-2316. Preservation-packaging shall be level A or C, and packing shall be level A, B, or C (see 6.2). Preservation-packaging and packing for commercial shipments shall be the same as that specified for level C.

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6. NOTES

6.1 Intended use. The cabinets covered herein are intended for use in kitchens and hospitals to keep food warm prior to serving.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type and style of cabinet required (see 1.2).
(c) Electrical characteristics, if other than specified (see 3.4.4).
(d) Selection of the applicable levels of preservation-packaging and packing (see 5.1).

6.3 Contract data requirements. Any data items to be delivered under any contract for the items covered by this specification should be specifically called for in the contract in accordance with the applicable regulations of the procuring activity.

6.4 Changes from previous issue. The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Navy - MC
Air Force - 99

Preparing activity:

Army - GL

Project No. 7310-0500

Review activities:

Army - MD
Navy - SA

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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

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